

LAB 3: BLOCK CIPHER OPERATIONS.

- INSTALLATIONS OF REQUIRED TOOLS

```
crypto@crypto:~$ sudo apt-get install openssl
[sudo] password for crypto:
Reading package lists... Done
Building dependency tree
Reading state information... Done
openssl is already the newest version (1.1.0g-2ubuntu4).
0 to upgrade, 0 to newly install, 0 to remove and 0 not to upgrade.
```

Openssl is a tool for TLS and SSL protocol.

```
crypto@crypto:~$ sudo apt-get install bless
Reading package lists... Done
Building dependency tree
Reading state information... Done
bless is already the newest version (0.6.0-5).
0 to upgrade, 0 to newly install, 0 to remove and 0 not to upgrade.
```

Bless is a tool used to view encrypted files in hexadecimal.

❖ TASK 1: ENCRYPTING USING DIFFERENT CIPHER MODES

```
crypto@crypto:~/Desktop$ openssl enc -aes-128-cbc -e -in plain.txt -out cipher.b
in -K 00112233445566778899aabbccddeeff -iv 0102030405060708
crypto@crypto:~/Desktop$
```



ENCRYPTION USING -aes-128-cbc

```
crypto@crypto:~/Desktop$ openssl enc -aes-128-cbc -d -out plain.txt -in cipher.bin -K 00112233445566778899aabbccddeeff -iv 0102030405060708
crypto@crypto:~/Desktop$
```

000aef0 c632 f2d7 fd3c 6cda 2991 f68e b5ff c4db
000af00

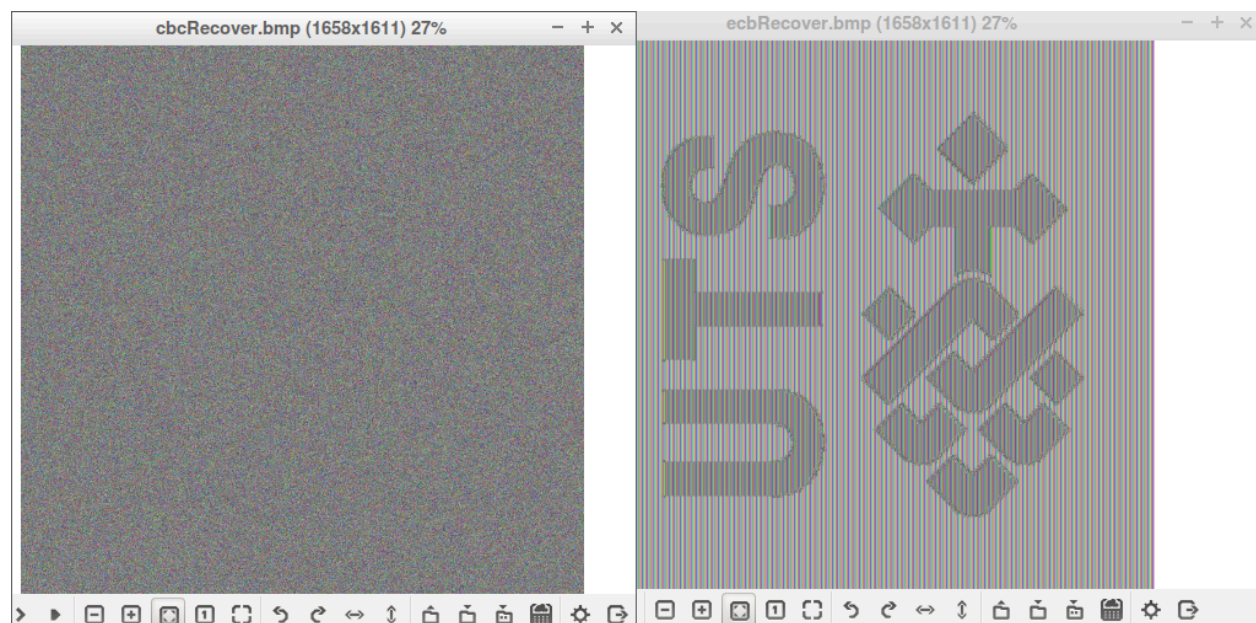
```
crypto@crypto:~/Desktop$ openssl enc -aes-128-cfb -e -in plain.txt -out cipher1.bin -K 00112233445566778899aabbccddeeff -iv 0102030405060708
crypto@crypto:~/Desktop$ openssl enc -aes-128-cfb -d -out plain.txt -in cipher1.bin -K 00112233445566778899aabbccddeeff -iv 0102030405060708
crypto@crypto:~/Desktop$ openssl enc -bf-cbc -e -in plain.txt -out cipher2.bin -K 00112233445566778899aabbccddeeff -iv 0102030405060708
crypto@crypto:~/Desktop$ openssl enc -bf-cbc -d -in cipher2.bin -K 00112233445566778899aabbccddeeff -iv 0102030405060708
```

Encryption using -aes-128-cfb and -bf-cbc

❖ TASK 2 : Encryption Mode – ECB vs CBC

```
crypto@crypto:~/Desktop$ openssl enc -aes-128-cbc -e -in uts.bmp -out cbc.bmp -K 001122334455667788aabbccddeeff -iv 12345678901234567890123456789012
crypto@crypto:~/Desktop$ openssl enc -aes-128-ecb -e -in uts.bmp -out ecb.bmp -K 001122334455667788aabbccddeeff -iv 12345678901234567890123456789012
warning: iv not use by this cipher
crypto@crypto:~/Desktop$ tail -c +55 ecb.bmp > ecbBody
crypto@crypto:~/Desktop$ tail -c +55 cbc.bmp > cbcBody
crypto@crypto:~/Desktop$ head -c 54 uts.bmp > header
crypto@crypto:~/Desktop$ cat header cbcBody>cbcRecover.bmp
crypto@crypto:~/Desktop$ cat header ecbBody>ecbRecover.bmp
```

Encrypting the.bmp file with 2 different ciphers

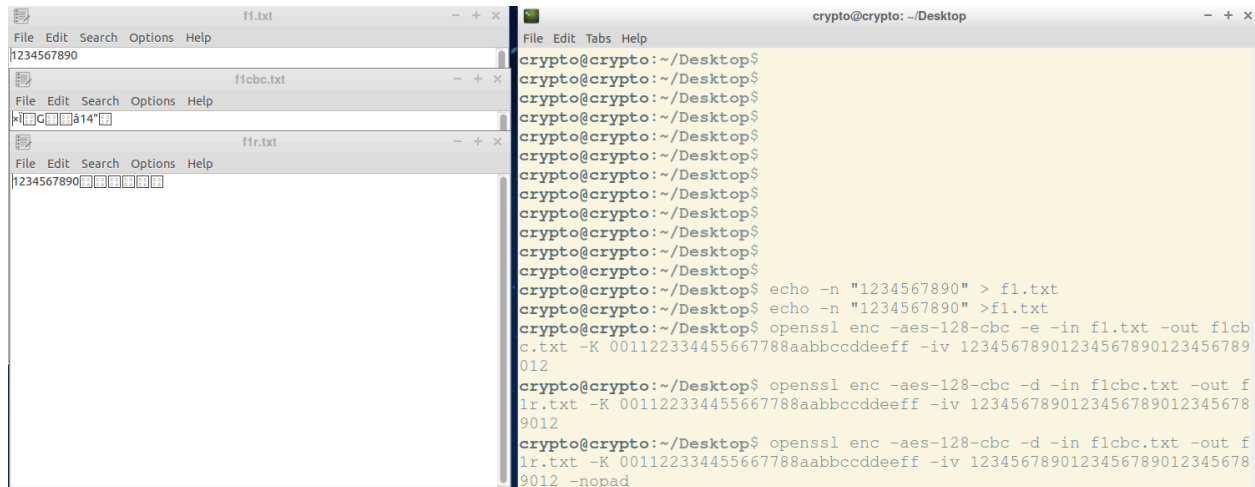


Recovered image of the encrypted .bmp files.

Q: Can you derive any useful information about the original picture from the encrypted picture?
Please explain your observations.

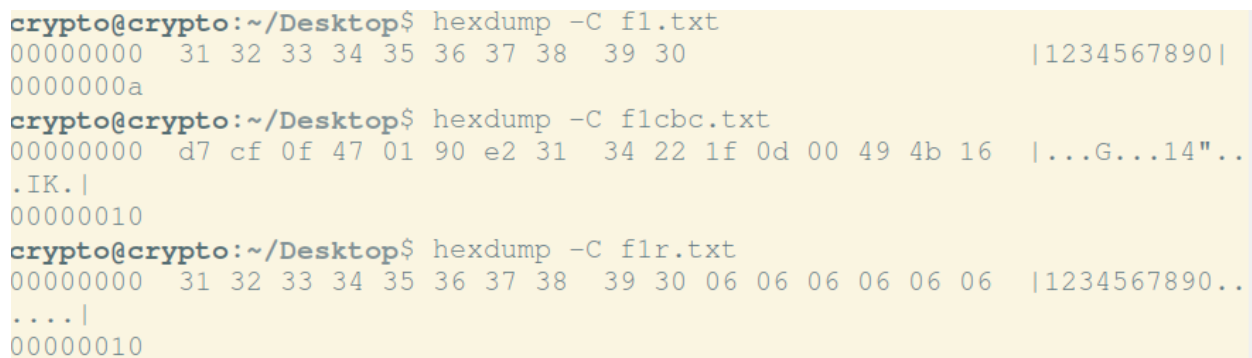
An original image was encrypted using two methods: AES-128 in CBC mode and AES-128 in ECB mode. When we attempted to recover the images, the CBC mode resulted in a completely gray image, while the ECB mode revealed a grayscale version of the original picture.

❖ TASK 3: PADDING



```
crypto@crypto: ~/Desktop
crypto@crypto:~/Desktop$ echo -n "1234567890" > f1.txt
crypto@crypto:~/Desktop$ echo -n "1234567890" > f1.txt
crypto@crypto:~/Desktop$ openssl enc -aes-128-cbc -e -in f1.txt -out f1cbc.txt -K 001122334455667788aabbccddeeff -iv 1234567890123456789012
crypto@crypto:~/Desktop$ openssl enc -aes-128-cbc -d -in f1cbc.txt -out f1r.txt -K 001122334455667788aabbccddeeff -iv 1234567890123456789012 -nopad
```

Encrypting a file to view the padding. Decrypting the file using -nopad command to view the padding added.



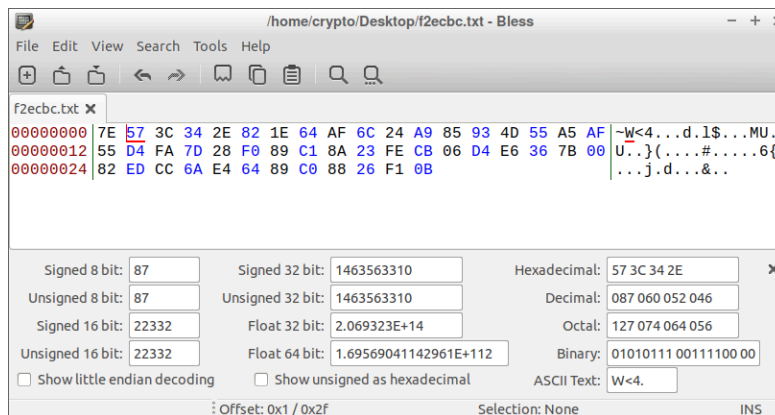
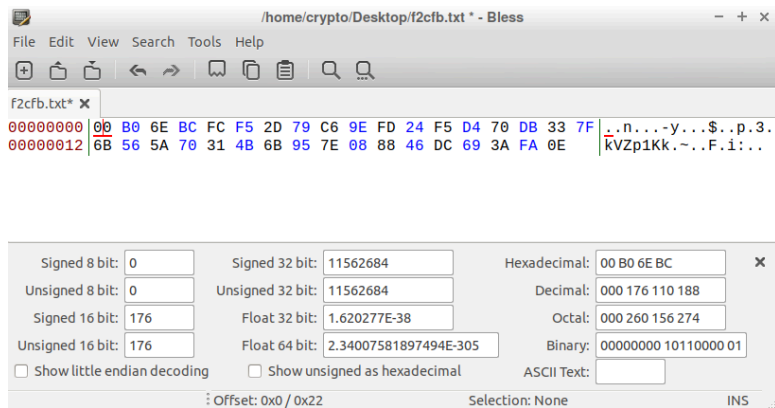
```
crypto@crypto:~/Desktop$ hexdump -C f1.txt
00000000  31 32 33 34 35 36 37 38  39 30                |1234567890|
0000000a

crypto@crypto:~/Desktop$ hexdump -C f1cbc.txt
00000000  d7 cf 0f 47 01 90 e2 31  34 22 1f 0d 00 49 4b 16  |...G...14"..
.IK.|
00000010

crypto@crypto:~/Desktop$ hexdump -C f1r.txt
00000000  31 32 33 34 35 36 37 38  39 30 06 06 06 06 06 06  |1234567890..
....|
00000010
```

Hexadecimal of the above files.

❖ TASK 4 ERROR PROPAGATION



Corruption using bless.

```

crypto@crypto:~/Desktop$ hexdump -C f2cfbr.txt
00000000 81 74 79 bd 66 55 ca 48 2d a0 32 27 dc b5 36 57 |.ty.fU.H-.2'..6W|
00000010 ee 38 31 32 33 34 35 36 37 38 31 32 33 34 35 36 |.812345678123456|
00000020 37 38 39 |789|
00000023
crypto@crypto:~/Desktop$ hexdump -C f2ecbr.txt
00000000 97 59 17 f2 10 4e 73 61 24 5a 35 8b a0 0d ed b8 |.Y...Nsa$Z5.....|
00000010 f3 5f 32 a7 b8 3c 31 56 b1 2c 7b 8a 6e c7 84 70 |. _2..<1V.,{.n..p|
00000020 02 ff 4f c2 f7 44 0b a5 88 4a 5c 62 77 f7 5a 21 |..O..D...J\bw.Z!|
00000030 49 fa 73 e3 87 40 b4 b6 d7 49 5b 78 d6 88 14 98 |I.s..@...I[x....|
00000040

```

Hexdump of the decrypted corrupted cipher using the key and iv.

Q: How much information can you recover by decrypting the corrupted file, if the encryption mode is ECB, CBC, CFB, OFB, and CTR respectively? Please find it out after you finish this task and provide justification.

❖ TASK 5 INITIAL VECTOR (IV)

```
crypto@crypto:~/Desktop$ hexdump -C f2cbciv2.txt
00000000  e6 e2 d9 e4 3c f6 56 aa 44 6b f4 18 7d 1f 64 f9 |....<.V.Dk..}.d.|
00000010  59 88 6b 53 f2 03 28 ad 7e e1 be d7 94 25 b2 6f |Y.kS..(.~....%.o|
00000020  ce 2d 3d e6 28 35 91 e7 68 04 f3 1d 73 51 d9 d7 |.-=.(5..h...sQ..|
00000030
crypto@crypto:~/Desktop$ hexdump -C f2cbciv.txt
00000000  7a f5 18 24 41 ed f9 19 52 a6 29 82 54 99 a1 22 |z..$A...R.).T.."|
00000010  d0 62 ab 49 b7 76 aa d3 b3 ae 79 25 8b da 46 9c |.b.I.v....y%..F.|
00000020  fd 14 da e7 e2 7e 28 ad 43 9b 4f fe 16 52 bf 8a |.....~(.C.O..R..|
00000030
```

Using different IV on the same plaintext file.

Q: What properties the IV should have? Why?

Uniqueness prevents predictability: By using a unique IV for each encryption, you ensure that the ciphertext doesn't reveal patterns based on repeated plaintext blocks.

Unpredictability thwarts certain attacks: If an attacker can predict the IV, they might be able to launch specific attacks against the encryption scheme.

Correct size ensures proper operation: Using the appropriate size for the IV ensures the cryptographic mode functions as intended and avoids potential vulnerabilities.

LAB SUMMARY:

1. Learnt how to use openssl
2. The various ciphers used in openssl
3. Understood the concept of padding
4. Importance and functions of iv was learnt.

References:

1. gemini ai was used for paraphrasing and forming meaningful sentences.

❖ I could't save these files on the path with my name due to storage issues. Please excuse me for the same.